Project 2: Turbojet Engine Design/Off-Design Performance Analysis

Riley J. Kyle*
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

Dr. David W. Riggins†
Missouri University of Science and Technology, Rolla, Missouri, 65409, United States

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

Nomenclature

T = temperature $[K]$
P = pressure $[Pa]$
ρ = density $\left[kg/m^3\right]$
T_t = total temperature $[K]$
P_t = total pressure $[Pa]$
τ = $\frac{T_{t,out}}{T_{t,in}}$
π = $\frac{P_{t,out}}{P_{t,in}}$
η = isentropic efficiency
u = velocity $[m/s]$
M = mach number
A = area $\left[m^2\right]$

Subscripts

0 = freestream
d = diffuser
c = compressor
b = burner
t = turbine
n = nozzle

*Aerospace Engineering Student, MAE
†Professor of Mechanical and Aerospace Engineering, MAE

I. Introduction

The purpose of this project is to plot the performance three space of a fixed-area missile-class engine with a simple converging nozzle. This will be done by varying the fuel flow rate from 20% to 500% the on design fuel rate, varying the Mach number from 0 to 2.5. Additionally, this will be done at sea level, 4000 m, 8000 m, and 12000 m altitude.

II. Completely define the on-design engine fluid dynamics and performance

A. Methodology

B. Results

Table 1 On-Design Operation

		Station						
		0	1	2	3	4	5	8
u	m/s	261.873	261.873	160.826	160.826	708.696	566.957	621.38
P	Pa	35651	35651	47679.6	758088	431148	171131	152490
T	K	236.23	236.23	257.491	604.339	1250	993.151	960.96
M		0.85	0.85	0.5	0.32637	1	0.897506	1
ρ	kg/m^3	0.525841	0.525841	0.645192	4.37076	1.20181	0.600386	0.55291
A	m^2	0.00318	0.00318	0.00422016	0.000622961	0.000514136	0.00128645	0.00127456
P_t	Pa	57177.7	57177.7	56558.1	816133	816133	288653	288653
T_t	K	270.365	270.365	270.365	617.214	1500	1153.15	1153.15

Table 2 On-Design Operation

τ_t		261.873
π_t		35651
F	N	236.23
$\dot{m}_{air}$	kg/s	0.85
$\dot{m}_{fuel}$	kg/s	0.525841
f		0.00318

III. Develop the complete performance envelope for this specific turbojet engine

A. Methodology

B. Results

1. Sea Level

2. 4000 m

3. 8000 m

4. 12000 m

IV. Conclusion

Appendix